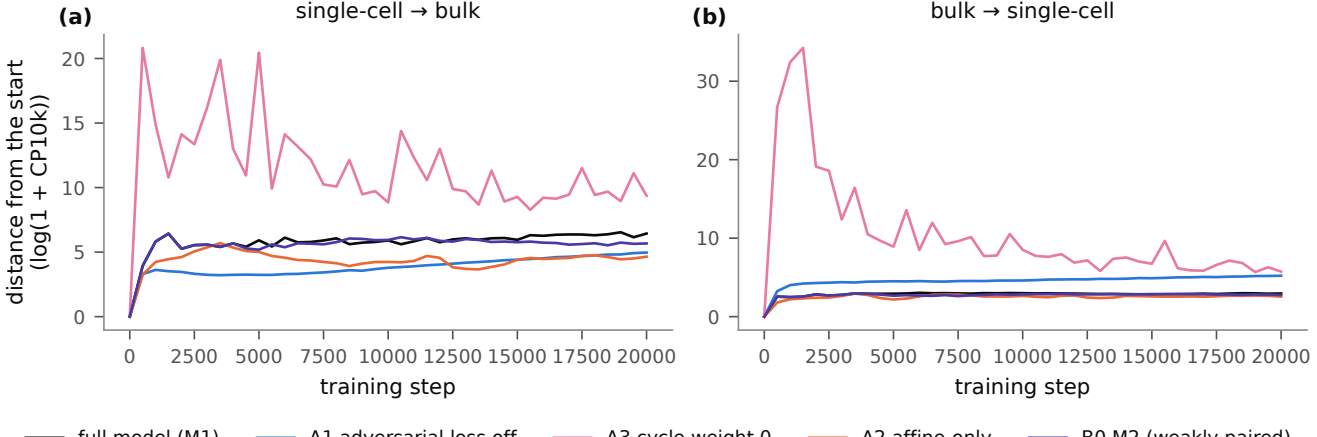
(a)
single-cell → bulk
distance from the start
(log(1 + CP10k))
0
5
10
15
20
0
2500
5000
7500
10000
12500
15000
17500
20000
training step
(b)
bulk → single-cell
0
10
20
30
0
2500
5000
7500
10000
12500
15000
17500
20000
training step
full model (M1)
A1 adversarial loss off
A3 cycle weight 0
A2 affine-only
B0 M2 (weakly paired)